Linear Programming

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1 Objective Function and Constraints

An objective function is an expression to maximize or minimize, for example

$$\boxed{Z = ax + by}.$$

Here x and y are decision variables. Constraints are inequalities such as

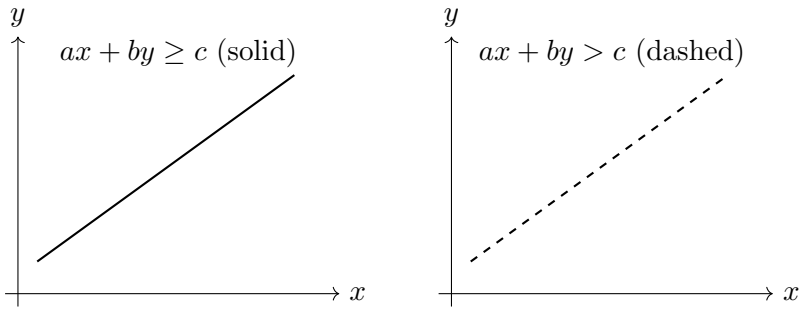
$$x \geq a, \qquad y \geq a, \qquad x + y \geq a.$$

2 Feasible and Infeasible Regions

The feasible region is the common region satisfying all constraints. Points outside it are infeasible. A bounded feasible region lies within a finite boundary. An unbounded feasible region extends indefinitely in at least one direction.

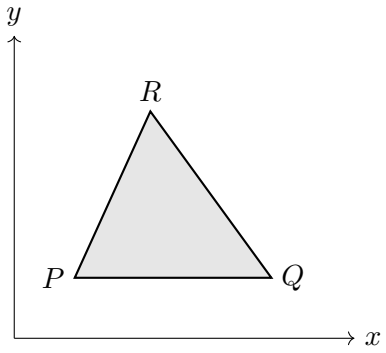
3 Drawing Inequalities

The boundary line $ax + by = c$ is drawn solid for $\geq$ or $\leq$ and dashed for $>$ or $<$.



4 Corner Points

Corner points are the vertices of the feasible region. They are candidates for maximum and minimum values of a linear objective function.



5 Finding Maximum and Minimum

1. Draw the constraints and identify the feasible region.

2. Find the corner points.
3. Substitute the coordinates of each corner point into $Z = ax + by$.
4. Compare the resulting values to find the maximum or minimum.

6 Important Notes

If two adjacent corner points give the same optimal value, every point on the joining segment is also optimal. For an unbounded region, corner point comparison alone may not establish the existence of an optimum. Examine the objective function over the unbounded directions. If the feasible region is empty, the problem has no feasible solution.